

# Analyzing NYPD Stop and Frisk Data in 2025

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## Introduction

The publicly available NYPD Stop, Question, and Frisk (SQF) dataset records police encounters in New York City dating back to 2003. Each record describes a single stop: when and where it occurred, the demographic characteristics of the suspect, the officer's stated reasons for suspicion, and whether the encounter resulted in a frisk or recovered weapon. A police frisk is a pat-down search of a suspect's outer clothing, legally permissible under *Terry v. Ohio* (1968), which established that officers may briefly detain and search individuals when they have reasonable suspicion of criminal activity and believe the suspect is armed and dangerous.

Stop-and-frisk has been the subject of sustained legal and public debate, particularly regarding whether the discretion afforded to officers produces racially patterned outcomes. *Floyd v. City of New York* (2013) found that the NYPD's stop-and-frisk practices had been conducted in a manner that violated the Fourth and Fourteenth Amendments, citing evidence of racial profiling. In the years since, the volume of stops has fallen dramatically, but questions about whether the remaining stops are conducted equitably have not gone away.

This analysis uses SQF data from 2025 to investigate one specific question: **among individuals who were stopped, does race predict whether they were frisked, and how does the answer change when we account for the officer's stated reasons for suspicion?** I take a three-part approach. First, I run a regression staircase that progressively adds controls and records how the racial gap in frisk rates evolves. Second, I examine whether the controls themselves, the officer-reported suspicion flags, are racially patterned. Third, I apply an outcome test (Knowles, Persico, and Todd, 2001) that compares weapon-recovery rates across races among frisked suspects, providing an independent check on whether officers apply consistent thresholds of suspicion across groups.

A note on what this analysis can and cannot show: the SQF dataset records only stops that occurred. My findings address what happens *after* a stop occurs, meaning the data is related to who is frisked, and whether those frisks recover weapons at similar rates across groups.

## TL;DR (Summary of Report)

This analysis proceeded in three steps to triangulate the central question from different angles. First, I calculated overall frisk rates by race and sex, finding substantial disparities: Black suspects represented the largest group by stop volume and were frisked at more than twice the rate of White suspects. Because raw gaps can reflect legitimate differences in stop circumstances rather than differential treatment, I then fit a series of logistic regression models that progressively added controls for demographic factors, stop location and suspected crime, and

officer-reported suspicion flags. The racial gap in frisk probability shrank dramatically as controls were added, falling from roughly 29 percentage points in the unadjusted comparison to 3 percentage points and statistical insignificance in the full model.

This finding does not establish the absence of racial bias. I show that the same controls that explain away the raw gap are themselves racially patterned: Black suspects are flagged for concealed weapon possession at roughly three times the rate of White suspects, and are stopped for suspected weapon possession at nearly three times the rate. "Controlling for" a racially patterned variable can absorb rather than explain the disparity it captures. To investigate further, I applied an outcome test on weapon recovery rates among frisked suspects. After accounting for stop context and stated suspicions, frisks of Black suspects recovered weapons 4.6 percentage points less often than frisks of White suspects ( $p = 0.006$ ), a pattern consistent with officers applying a lower threshold of suspicion when deciding to frisk Black suspects. Taken together, the three analyses suggest that racial disparities in this dataset may operate less at the moment of the frisk decision itself than at the upstream decision of when to characterize a suspect as potentially armed.

## **The Data**

The dataset is a publicly available Microsoft Excel file from NYC OpenData containing 26,422 rows, each representing one police stop conducted in 2025. Each row records the time, date, and location of the stop, the rank and command of the officer conducting it, the officer's stated suspicions, demographic information about the suspect, and outcomes including whether the stop resulted in a frisk and whether any weapons were recovered.

My primary outcome for the first two parts of the analysis is `FRISKED_FLAG`, which records whether the stop resulted in a pat-down search. For the third part, I used `WEAPON_FOUND_FLAG` to evaluate whether frisks successfully recovered a concealed weapon. The treatment variable throughout is `SUSPECT_RACE_DESCRIPTION`, which records the officer's perception of the suspect's race rather than a self-reported value. I retained all seven racial categories present in the data (White, Black, Black Hispanic, White Hispanic, Asian/Pacific Islander, American Indian/Alaskan Native, Middle Eastern/Southwest Asian) and set White suspects as the reference category. My demographic and contextual controls include `SUSPECT_SEX`, `SUSPECT_REPORTED_AGE`, `STOP_LOCATION_BORO_NAME`, and `SUSPECTED_CRIME_DESCRIPTION`. To capture the officer's stated reasons for suspicion, I also included a set of binary flag variables describing observed suspect actions (such as casing a location or concealed weapon possession) and background circumstances (such as a known history of violent crime or a known reputation for carrying weapons).

Data preparation required several steps. The flag variables in the raw data are stored as "Y" and "N" string values, which I converted to binary 0/1 columns for use in regression. I converted SUSPECT\_REPORTED\_AGE to numeric, dropping rows where age was missing or non-numeric. For the hit rate regression, I additionally filtered out suspected crime categories with fewer than 50 frisks or fewer than 5 weapon recoveries, since these categories produced perfect separation and prevented model convergence. After these steps, the frisk regression used 26,414 stops and the hit rate regression used 13,610 frisks.

Several variables in the dataset would have appeared analytically useful but were deliberately excluded. Post-frisk outcomes such as SEARCHED\_FLAG, ARREST\_FLAG, and the various PHYSICAL\_FORCE\_\* flags occur after the frisk decision and would introduce reverse causality if used as predictors. I also did not use Demeanor\_of\_Person\_Stopped, since this field is recorded by the same officer making the frisk decision and could itself reflect bias rather than describing suspect behavior objectively. The Denial\_Reason fields and similar variables are only populated for specific outcomes, making them inappropriate as controls.

## Methods

I used a three-part analytical strategy to investigate the question from multiple angles. The first part establishes a descriptive baseline by calculating frisk rates across demographic and contextual groups. The second part fits a series of logistic regression models to test whether the racial gaps observed in the descriptive analysis survive after accounting for legitimate factors. The third part applies an outcome test, comparing weapon recovery rates among frisked suspects, to provide an independent check on whether officers apply consistent thresholds of suspicion across racial groups. Each part has limitations that the others partially address, which is why I used all three together. I organized my Jupyter Notebook into clearly labeled sections corresponding to these three parts, with Markdown cells interpreting the results at each step.

In the first part, I calculated frisk rates using pandas across several groupings: race, sex, borough, and suspected crime category. This surface-level analysis was designed not to draw conclusions but to identify the patterns worth pursuing in the regression analysis that followed. The descriptive results raised the central question of the regression: do the racial gaps in frisk rates reflect differences in stop circumstances, or do they persist even within similar contexts?

In the regression section, I used logistic regression because the outcome variable (FRISKED\_FLAG) is binary. Each coefficient estimates the conditional association between race and frisk probability, holding the included controls constant. I used a "staircase" approach, fitting four nested models that progressively added controls. The baseline model included race only. The second model added demographic controls (sex and age). The third added the context of the

stop (borough and suspected crime). The fourth and final model added officer-stated suspicion flags and background circumstances. I reported marginal effects (percentage point changes) rather than raw log-odds coefficients because they are more interpretable, and I used White suspects as the reference category so that all race coefficients describe gaps relative to White suspects.

The regression results revealed that the strongest predictors of being frisked were the background circumstance flags (known to carry weapons, history of violent crime). Before moving to the third part of the analysis, I conducted a brief audit to test whether these controls were themselves racially patterned. If a control variable is applied differently across racial groups, then "controlling for" it in a regression can absorb rather than explain the underlying disparity. For each suspicion flag and for the most common crime categories, I computed the rate at which each variable was applied to stops of each racial group, along with ratios relative to White suspects, to summarize the differences.

In the third part, I applied a hit rate test based on the outcome test logic formalized by Knowles, Persico, and Todd (2001). The intuition: if officers apply a race-neutral threshold of suspicion when deciding to frisk, weapon recovery rates should be roughly equal across racial groups. A lower recovery rate for a particular group is consistent with officers having frisked that group on weaker evidence, while a higher rate is consistent with a stricter threshold. Among frisked suspects, I calculated weapon recovery rates by race and tested whether the rates differed using a chi-square test of independence. I ran a two-sample test of proportions for the Black-White comparison specifically. I then fit a logistic regression of WEAPON\_FOUND\_FLAG on race plus the same controls used in Part 2, to estimate the racial gap in recovery rates after adjusting for stop context. The simple outcome test has a known limitation called the infra-marginality problem, in which average hit rates can mislead when the underlying distributions of suspicion differ across groups. More rigorous threshold tests address this but require structural assumptions beyond the scope of this analysis, so the results should be read as suggestive rather than definitive.

# Findings

## Part 1

Of the 26,422 stops in the 2025 SQF data, 13,780 (52.2%) resulted in a frisk. Frisk rates varied substantially by suspect race. Black suspects had the highest rate at 56.9% across 15,538 stops, followed by Black Hispanic suspects at 55.9% (n = 2,614), American Indian/Alaskan Native suspects at 53.1% (n = 81), and White Hispanic suspects at 47.2% (n = 5,602). Middle Eastern/Southwest Asian suspects were frisked in 40.6% of stops (n = 313). At the low end, Asian/Pacific Islander suspects were frisked at 33.6% (n = 590), and White suspects at 28.0% (n = 1,692). The unadjusted gap between Black and White suspects is 29 percentage points, and the gap between Asian and White suspects is the only one where a non-White group has a lower rate than White suspects.

The pattern persists when the data is separated by sex. Black male suspects accounted for 14,287 of all stops, the largest single race-by-sex group in the data, and were frisked in 59.7% of those stops. White male suspects were frisked in 30.4% of 1,486 stops, a 29-percentage-point gap that mirrors the overall comparison. A small share of stops record sex as unspecified, which I include in the descriptive table for completeness but set aside for the male-female comparison. Frisk rates for female suspects are substantially lower across all racial groups, consistent with the broader pattern in the data that male suspects are far more likely to be frisked.

These descriptive findings establish that the raw racial gaps in frisk rates are large and persist within sex categories. They do not, however, tell us why the gaps exist. Black suspects in this dataset are stopped in different contexts than White suspects, for different suspected crimes, and with different stated reasons for suspicion. The next part of the analysis tests whether the racial gaps in frisk rates survive after accounting for these factors.

## Part 2

### Regression Staircase

Using the staircase method with logistic regression, I found that the racial gap in frisk probability shrinks dramatically as controls are added. The baseline model, which included race only, reported that Black suspects had the highest frisk probability at roughly 29 percentage points above White suspects. Adding sex and age reduced this to 21.1 percentage points. Adding borough location and suspected crime description shrank the Black coefficient further to 3.3 percentage points, and adding background circumstances and suspicion flags brought it to 1.6 percentage points. In the full model, the Black-White gap is small and no longer statistically significant.

Examining where the shrinkage occurred is itself informative. Adding stop location and suspected crime accounted for the largest single reduction in the racial gap, though the Black coefficient remained statistically significant at that stage. It was the addition of background circumstances and suspicion flags that shrank the coefficient to 1.6 percentage points and pushed it below statistical significance. This pattern suggests that most of the raw gap is associated with which stops happen to whom, rather than differential treatment of similar suspects within the same context.

The other racial groups followed a similar pattern. Adding controls also shrank the coefficients for Black Hispanic and White Hispanic suspects substantially, with both becoming statistically insignificant in the full model. The one exception is American Indian/Alaskan Native suspects, whose coefficient remained statistically significant across all four models, ending at roughly 8.2 percentage points in the full model. However, this group had only 81 stops in the data, so the estimate is imprecise and should not be over-read.

### **Auditing the Controls**

The regression staircase appears to deliver a tidy result: once stop circumstances and stated suspicions are accounted for, the racial gap in frisk rates nearly disappears. This conclusion, however, rests on an assumption worth testing directly. A regression that "controls for" a variable treats that variable as a legitimate, racially neutral input to the decision being studied. If officers instead apply those controls differently across racial groups, then controlling for them absorbs the disparity rather than explaining it.

To test this, I computed the rate at which each suspicion flag was applied to stops of each racial group. The differences are substantial, particularly for weapon-related flags. Black suspects were flagged for concealed weapon possession in 45.6% of stops, compared with 15.2% for White suspects, a striking difference. Black Hispanic suspects showed a similar elevation at 41.7%. Background circumstance flags showed smaller but still meaningful gaps, with Black and Hispanic groups flagged for violent crime history or known weapon carrying at roughly 1.5 times the White rate. Other flags, such as casing a location or drug transactions, did not show the same racial pattern, and in the case of the casing flag White suspects had the highest rate.

The distribution of suspected crimes followed the same pattern. Stops of Black suspects were categorized as criminal possession of a weapon (CPW) in 54.0% of cases, compared with 19.9% for White suspects. White suspects, by contrast, were more likely to be stopped for property crimes such as burglary. This matters because CPW and the concealed weapon possession flag were the two strongest predictors of being frisked in the regression staircase. The variables doing the most work to shrink the racial coefficient are precisely the variables that are most racially patterned.

This finding does not establish that officers are biased in how they assign suspicion. The data cannot distinguish whether the gaps reflect differences in suspect behavior, differences in officer perception, or some combination of both. What it does show is that the staircase result cannot be read as evidence that frisk decisions are race-neutral. The dramatic shrinkage of the racial coefficient does not resolve the question of bias so much as relocate it: from how officers act on suspicion, to how suspicion is characterized in the first place.

### **Part 3**

In Part 2, the strongest predictors of being frisked were the background circumstance flags: a suspect flagged for a violent crime history (log-odds 4.67,  $p < .01$ ) or known to carry a weapon (3.39,  $p < .01$ ). These flags drove much of the shrinkage in the racial coefficient, which raises a natural question. If officers are acting on consistent, race-neutral thresholds of suspicion, the frisks that result should succeed at similar rates across racial groups. This motivates the hit rate test: among frisked suspects, do weapon recovery rates differ by race?

Across the dataset, there were 13,780 frisks and 2,221 weapons recovered, an overall recovery rate of 16.1%. Separating by race reveals a clear gap. White suspects had a weapon recovery rate of 19.7% ( $n = 473$ ), compared with 15.4% for Black suspects ( $n = 8,843$ ). The unadjusted Black-White gap is 4.3 percentage points. A chi-square test confirmed that recovery rates differ significantly across racial groups ( $p = 0.0026$ ), and a two-sample test of proportions for the Black-White comparison specifically returned a difference of 4.3 percentage points ( $p = 0.012$ ).

Because the unadjusted gap could reflect differences in the contexts in which frisks occurred, I fit a logistic regression of weapon recovery on race plus the same controls used in Part 2. After adjustment, the gap did not diminish. Frisks of Black suspects recovered weapons 4.6 percentage points less often than frisks of White suspects ( $p = 0.006$ ), and frisks of Black Hispanic suspects 3.8 percentage points less often ( $p = 0.044$ ). This is the key contrast with Part 2: there, adding controls caused the racial gap in frisk rates to shrink toward zero. Here, adding the same controls left the racial gap in recovery rates essentially intact. The convergence adjustment described in the Methods section, which filtered out rare crime categories, did not meaningfully change the race coefficients, suggesting the result is robust to that specification choice.

Interpreted through the outcome test, this pattern is consistent with officers applying a lower threshold of suspicion when deciding to frisk Black and Black Hispanic suspects. A larger share of those frisks recovered nothing, which is what would be expected if the decision to frisk was made on weaker evidence. It is important to be precise about what this does and does not show. The test is consistent with a lower threshold; it does not prove discriminatory intent, and the data cannot rule out other explanations.



## Synthesis

Part 1 set the foundation for the rest of the analysis. The initial frisk rates revealed a large racial gap, a pattern that held when the data was disaggregated by sex. But these descriptive calculations were never going to be enough to support substantive conclusions, because raw rates conflate too many factors. Part 2 appeared to explain the gap away once controls were added, but the audit of those controls showed that they are themselves racially patterned. The gap was relocated, not resolved. Part 3 then identified a different gap, in weapon recovery rates, that the same controls did not explain away.

What makes these results worth the deeper analysis is the distance between the surface-level story and what the regression and outcome tests reveal. In Part 1, the frisk rates tell a simple story: large racial gaps in who is stopped and who is frisked. The regression staircase complicates that story. As controls were added, the racial coefficients diminished, but this is not evidence of an absence of racial bias. The controls that did the most to shrink the gap were the suspicion flags and suspected crime categories, and the audit showed those very variables to be strongly racially patterned. The evidence is therefore consistent with racial disparity operating less at the frisk decision itself and more at the upstream characterization of suspicion: which behaviors are flagged, which crimes are suspected, and how a suspect's background or appearance factors into that judgment. The hit rate test points in the same direction. Frisks of Black and Black Hispanic suspects recovered weapons less often even after adjustment, which is consistent with those frisks having been initiated on weaker evidence. Both findings, the differential flagging and the lower recovery rate, point at the same stage of the process: the formation of suspicion rather than the mechanics of the frisk.

The value of this distinction is that it is more specific, and more useful, than a verdict of "discrimination" or "no discrimination." A raw comparison of frisk rates cannot tell you where in the policing process a disparity originates, and the regression alone, read carelessly, could even be taken to suggest the disparity is not real. Taken together, the three analyses suggest that scrutiny of stop-and-frisk practices should focus less on the frisk decision in isolation and more on how suspicion is formed and recorded in the first place. An intervention aimed only at the moment of the frisk would, if this interpretation is correct, miss the stage where the disparity actually concentrates.

## Limitations

Two limitations meaningfully constrain how this analysis should be read. The first is the selection problem: the dataset records only stops that occurred. There is no record of the pedestrians officers observed but chose not to stop. This means the analysis cannot speak to

whether the stop decision itself is racially patterned, since every row in the data represents a stop that already happened. Part 3 brings the analysis closer to officer motivation by examining outcomes among frisked suspects, and the finding that frisks of Black and Black Hispanic suspects recovered weapons less often is consistent with a lower threshold of suspicion. Even so, this finding still describes only the stops and frisks that took place, not the broader population of encounters officers could have initiated.

The second limitation is that nearly every relevant variable in the dataset, including race, suspicion flags, and suspected crime, is recorded by the same officer making the frisk decision. The controls in the regression therefore reflect the officer's framing of the encounter rather than an independent account of it. The audit in Part 2 showed those controls to be racially patterned, but the data alone cannot determine whether that pattern reflects differences in suspect behavior, differences in officer perception, or some combination of both. Resolving this question would require evidence from outside the SQF dataset, such as independent observation of encounters.

## **Conclusion**

This analysis found that raw racial gaps in frisk rates largely disappear once stop context is controlled for, but those controls themselves are racially patterned. The frisk rates of Black suspects and Black Hispanic suspects are shown to recover weapons less often even after adjustment, and they are found to be frisked more often than other racial groups. The disparity in this dataset is consistent with operating at the formation of suspicion, not at the mechanics of the frisks themselves. Looking at the number of frisks alone is not enough to draw conclusions about racial disparity or discrimination towards racial groups, but investigating the context behind the stop and the racial patterns within background characteristics proves to be an important analysis on its own. As mentioned in the limitations, these findings are bound to this dataset, meaning that the data is only consistent with stops that were reported, there is no confirmed correlation between these racial patterns and stops that did not occur.

Considering that I only analyzed data from 2025, a multi-year analysis would extend these findings and perhaps provide a bit more credibility or insight on police behavior and perceptions of stopped suspects over the years. New York City is a large and diverse city filled with history as well as political unrest, which may have an impact on some of the racial patterns. If this data were present in other major cities, I would be keen to test whether these results remain consistent in other cities.

## References

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